Gokul Vishnu

Bangalore, India (Open to relocate to Sweden)

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Summary

RTL Design Engineer with 5+ years of experience in FPGA development across industrial and aerospace domains. Strong expertise in Verilog- and VHDL-based digital design, protocol implementation, and timing-driven architecture for high-integrity systems. Proven experience in AXI-based designs, DDR data paths, and image sensor interfaces from concept through hardware validation. Skilled in simulation, synthesis, and lab bring-up using industry-standard FPGA toolchains. Recognized for cross-functional collaboration and structured documentation practices including Conceptual Design Documents (CDD), Module Requirements Documents (MRD), and Interface Control Documents (ICD) preparation to ensure precise and compliant design specifications.

Core Skills

- **Digital Design:** RTL design, FPGA RTL
- **HDLs & Programming:** Verilog HDL, VHDL
- **Protocols & Buses:** AXI Stream, AXI Lite, AXI Full, MIL 1553B, SPI, UART, PCI
- **FPGA Tools:** Xilinx Vivado, ModelSim, Libero, Lattice Diamond
- **Devices & Interfaces:** Xilinx xc7z030fbg676-1, Lattice LFE2 20E, Microchip M2GL090T(S)-FG676
- **Methods & Docs:** CDD, MRD, ICD

Experience

Klok Systems India Pvt. Ltd. **August 2023 – Present**

Senior Engineer *Bangalore, India*

- Own RTL design for FPGA modules, aligning protocol requirements with timing, area and interface goals.
- Develop RTL for Device DMA, MIL-1553B and Device Interface blocks, integrating cleanly with system-level buses.
- Support simulation of PCI Interface RTL for functional checks and refined code using coverage report analysis.
- Run synthesis to review warnings, updated RTL to close issues and maintained CDD, MRD and ICD design documents.

Capgemini Engineering **January 2023 – June 2023**

Software Engineer Lead *Bangalore, India*

- Simulated RTL behavior against Module Requirements Documents to confirm logic and clock margin targets.
- Performed coverage report analysis for complex RTL blocks, closing gaps and improving verification quality metrics.
- Supported source code reviews against MRD, ensuring RTL implementation met functional and quality requirements.
- Worked with cross-functional design and verification teams to align RTL updates with system integration milestones.

Swan Sorter Systems Pvt. Ltd. **December 2020 – January 2023**

Hardware Design Engineer *Bangalore, India*

- Designed RTL for color sorting algorithm modules on Xilinx FPGAs to support high-throughput inspection.
- Developed DDR data acquisition with AXI DMA to move sensor streams while minimizing processor load.
- Programmed Toshiba image sensors and validated data paths from sensor to memory and sorting logic.
- Delivered FPGA designs and integrations for hardware testing while meeting schedule and quality targets.

Projects

CDU (Control Display Unit) - Collins Aerospace (Client) **January 2026 – Present**

- Owned CDD and MRD development for Keypad2, Keypad4 and Datamux modules.
- Implemented RTL design for Keypad and Datamux modules.
- Resolved source code and MRD issues to ensure stable module integration.

GenericPCI Bridge – Collins Aerospace (Client) **February 2024 – December 2025**

- Owned CDD/MRD for Device DMA, Device Interface, and PCI Interface (memory-mapped and burst modes).
- Implemented RTL for Device Interface and Device DMA (memory-mapped and burst mode paths).
- Developed CDD, MRD and RTL for Device Bus arbitration, fixed PCI Interface Custom Source Code and MRD issues.

MIL 1553B Protocol – Klok Systems **August 2023 – January 2024**

- Authored the Interface Control Document (ICD) defining RT register map, MIL-1553B interface signals and timing.
- Prepared the Module Requirements Document capturing RT behaviour and protocol flows.
- Designed RTL for Manchester encoder/decoder plus logic to split BC command and mode-code words into address fields.

MRD and Source Code Support – Capgemini Engineering **February 2023 – April 2023**

- Simulated RTL modules against MRD to confirm functional intent and clock domain constraints.
- Performed coverage report analysis for complex RTL designs, identifying gaps and driving additional test scenarios.
- Supported source code reviews against MRD, ensuring RTL implementation met functional and quality requirements.

Color Sorting Algorithm – Swan Sorter Systems Pvt. Ltd. **May 2022 – December 2022**

- Designed RTL to receive HMI data and apply flat-line correction on incoming sensor pixels for accurate processing.
- Implemented RTL to convert RGB inputs to HSL, enabling reliable color classification in the sorting workflow.

Data Acquisition to DDR Memory – Swan Sorter Systems Pvt. Ltd. **July 2021 – April 2022**

- Developed AXI DMA based block design to stream high-speed sensor data reliably into DDR memory buffers.
- Created custom RTL to synchronize Zynq processor and FPGA domains, ensuring coherent, low-latency data transfers.

Sensor Programming – Swan Sorter Systems Pvt. Ltd. **December 2020 – May 2021**

- Programmed Toshiba TCD2564DG and TCD1209DG image sensors to extract stable and usable image data.
- Coordinated with hardware teams to tune sensor settings for noise, exposure, and data integrity.

Key Achievements

- Completed complex RTL design ahead of schedule, improving manager satisfaction and driving repeat business.
- Achieved full pass rate in hardware validation for owned design modules against project requirements.
- Delivered key milestones under budget while improving design efficiency and overall functionality.

Education

East Point College of Engineering and Technology **August 2015 – June 2019**

Bachelor of Engineering in Electronics and Communication *Bangalore, India*

Certifications

- Professional Development Course in VLSI Design, Sandeepani School of VLSI Design, 2020.

Languages

- English – Full Professional Proficiency (C1)
- Swedish – Basic Proficiency (A1)